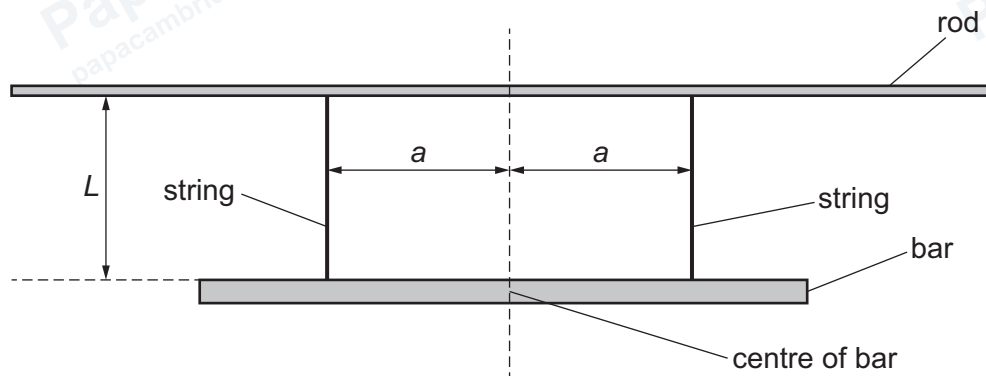


- 1 A rod is fixed horizontally. A solid bar of mass  $M$  is attached to the rod by two vertical strings so that the bar is horizontal, as shown in Figure 1.1.



**Figure 1.1**

The vertical strings are equal horizontal distances  $a$  from the centre of the bar.

The separation between the rod and the bar is  $L$ .

One end of the bar is displaced horizontally and then released. The bar rotates with oscillations about its centre. The period of the oscillations is  $T$ .

It is suggested that  $T$  is related to  $a$  by the relationship

$$T = \frac{2\pi P}{a^q} \sqrt{\frac{L}{M}}$$

where  $P$  and  $q$  are constants.

Plan a laboratory experiment to test the relationship between  $T$  and  $a$ .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for  $P$  and  $q$ .

In your plan, you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

## Diagram

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